

assignment9 - cyclotron motion

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:13pm

Quiz Instructions

Will make a tutorial

2 attempts



Question 1 10 pts

A particle ($q = 5.0 \text{ nC}$, $m = 3.0 \mu\text{g}$) moves in a region where the magnetic field has components $B_x = 2.0 \text{ mT}$, $B_y = 3.0 \text{ mT}$, and $B_z = -4.0 \text{ mT}$. At an instant when the speed of the particle is 5.0 km/s and the direction of its velocity is 120° relative to the magnetic field, what is the magnitude of the acceleration of the particle?

hint: This is not hard. magnetic force = $qvB\sin(\text{angle between B and v})$.

There is a tutorial



45 m/s/s



39 m/s/s



25 m/s/s



17 m/s/s

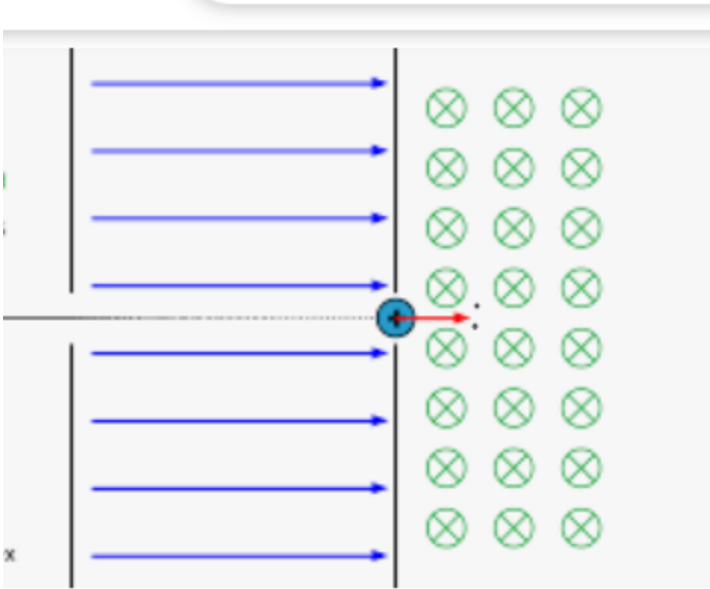


Question 2 10 pts

A proton is accelerated from rest through 0.50 kV . It then enters a uniform magnetic field of 0.30 T that is oriented perpendicular to its direction of motion.

(a) What is the radius of the path the proton follows in the magnetic field?

(b) How long does it take the proton to make one complete circle in the magnetic field?.



There is a tutorial. Use conservation of energy to find the speed. $eV = KE$ then slide 61 for $r = (mv)/(qB)$

□

○

(a) 11 mm (b) 0.22 us

○

(a) 1m (b) 1 s

○

(a) 20 mm (b) 0.22 s

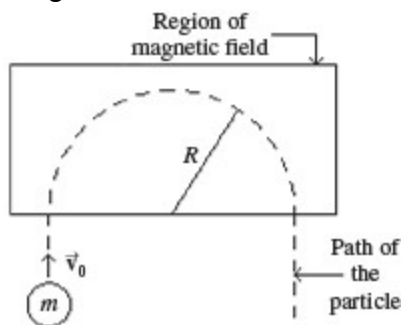
○

(a) 10 cm (b) 20 ms

⋮

Question 3 10 pts

In the figure, a small particle of charge $-1.9 \times 10^{-6} \text{ C}$ and mass $3.1 \times 10^{-12} \text{ kg}$ has speed $V = 8.1 \times 10^3 \text{ m/s}$ as it enters a region of uniform magnetic field. The particle is initially traveling perpendicular to the magnetic field and is observed to travel in the semicircular path shown with radius $R = 5.0 \text{ cm}$. Find the magnitude and direction of the magnetic field in the region.



hint: slide 61 has the equation for the radius.

☐

0T

☐

0.56T toward you

☐

0.26T into the paper

☐

0.56 T into the paper

☐

0.26T toward you

☐

Question 4 10 pts

A proton having a speed of 3.0×10^6 m/s in a direction perpendicular to a uniform magnetic field moves in a circle of radius 0.20 m within the field. What is the magnitude of the magnetic field? ($e = 1.60 \times 10^{-19}$ C, $m_{\text{proton}} = 1.67 \times 10^{-27}$ kg)

☐

0.080 T

☐

0.16 T

☐

0.24 T

☐

0.36 T

☐

0.32 T

☐

Question 5 10 pts

An electron is accelerated from rest through a potential difference of 3.75 kV. It enters a region where a uniform 4.0-mT magnetic field is perpendicular to the velocity of the electron. Calculate the radius of the path this electron will follow in the magnetic field. ($e = 1.60 \times 10^{-19}$ C, $m_{\text{electron}} = 9.11 \times 10^{-31}$ kg)

☐

4.2 cm

☐

1.2 cm

☐

2.2 cm

☐

3.2 cm

☐

5.2 cm

☐

Question 6 10 pts

A doubly charged ion with speed 6.9×10^6 m/s enters a uniform 0.80-T magnetic field, traveling perpendicular to the field. Once in the field, it moves in a circular arc of radius 30 cm. What is the mass of this ion? ($e = 1.60 \times 10^{-19}$ C)

☐

11×10^{-27} kg

☐

8.2×10^{-27} kg

☐

3.3×10^{-27} kg

☐

6.7×10^{-27} kg

☐

Question 7 10 pts

Alpha particles, each having a charge of $+2e$ and a mass of 6.64×10^{-27} kg, are accelerated in a uniform 0.80-T magnetic field to a final orbit radius of 0.30 m. The field is perpendicular to the velocity of the particles. How long does it take an alpha particle to make one complete circle in the final orbit? ($e = 1.60 \times 10^{-19}$ C)

Slide 61. The period is $(2\pi)(m)/(qB)$

☐

0.25 μ s

☐

0.49 μ s

☐

0.16 μ s

☐

0.33 μ s

☐

0.40 μ s

☐

Question 8 10 pts

Alpha particles, each having a charge of $+2e$ and a mass of 6.64×10^{-27} kg, are accelerated in a uniform 0.5T magnetic field to a final orbit radius of 0.5 m The field is perpendicular to the velocity of the particles. What is the kinetic energy of an alpha particle in the final orbit? ($1 \text{ eV} = 1.60 \times 10^{-19}$ J, $e = 1.60 \times 10^{-19}$ C)

hint: The KE is $0.5 m v^2$. If you have r , you can find the speed. then the KE. divided by e to convert J into eV. M is mega or one million.

☐

2.6 MeV

☐

3.9 MeV

☐

3.4 MeV

☐

4.3 MeV

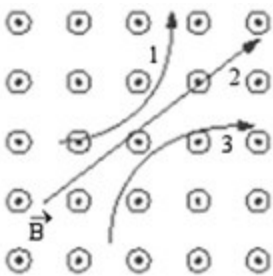
☐

3.0 MeV

☒

Question 9 10 pts

Three particles travel through a region of space where the magnetic field is out of the page, as shown in the figure. What are the signs of the charges of these three particles?


☐

1 is neutral, 2 is positive, and 3 is negative.

☐

1 is neutral, 2 is negative, and 3 is positive.

☐

1 is positive, 2 is negative, and 3 is neutral.

☐

1 is negative, 2 is neutral, and 3 is positive.

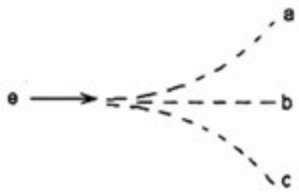
☐

1 is positive, 2 is neutral, and 3 is negative.

☒

Question 10 10 pts

An electron is moving to the right, as shown in the figure. Suddenly it encounters uniform magnetic field pointing out of the page. Which one of the three paths shown will it follow in the field?

☐

path b

☐

path a

☐

path c

Not saved

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